

Module – 4

Role of Signs and Markings in Safety

Introduction:

Road signs and markings form the silent language of roads. They communicate essential information to drivers and pedestrians without words. In traffic control, they serve as preventive, regulatory, and informative tools.

Signs: Visual boards placed along the road to convey rules, warnings, or guidance.

Markings: Painted lines and symbols on the road surface guiding lane use, pedestrian movement, and turning rules.

Example

- A STOP sign at an intersection prevents side collisions.
- Zebra crossings with markings near schools reduce pedestrian crashes.

Types of Road Signs

In India, road signs are classified into three major categories: Regulatory, Warning, and Informatory signs. Each category uses specific shapes and colors, chosen scientifically to ensure uniformity and instant recognition.

a) *Regulatory Signs*

Regulatory signs provide mandatory instructions that enforce road discipline. By clearly stating what is permitted or prohibited, they bring uniformity in driver actions and prevent unsafe behavior.

- **Shape & Color:** Mostly circular with white background, red border, and black symbols/text.
- **Special Cases:**
 - **STOP:** Octagonal red sign with white letters.
 - **GIVE WAY:** Inverted triangle with red border and white background.
- **Examples:** No Entry, Speed Limit, No Parking, No Overtaking.
- **Role in Safety:** Prevents violations and enforces discipline. For example, a “No Overtaking” sign on a narrow bridge avoids head-on collisions.

b) Warning Signs

Warning signs act as advance alerts. They warn drivers of hazards or unusual conditions ahead, giving them enough time to react safely.

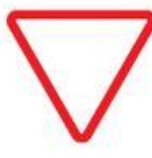
- Shape & Color: Equilateral triangle with red border, white background, and black symbols.
- Examples: Curve Ahead, School Ahead, Slippery Road, Pedestrian Crossing, Narrow Bridge.
- Role in Safety: By giving advance notice, drivers slow down and take precautions. This reduces the chances of surprise reactions, which are often the cause of crashes. For example, a “School Ahead” sign reminds drivers to be alert for children crossing.

c) Informatory / Guide Signs

Guide or informatory signs provide clarity and reduce confusion, especially for drivers who are unfamiliar with a route. By giving directions and service information, they minimize unsafe last-minute decisions such as sudden lane changes or abrupt stops.

- Shape & Color: Rectangular.
 - Blue background with white symbols/text → Services (Hospital, Parking, Fuel Station).
 - Green background with white text → Direction, route, and distance signs.
- Examples: Bus Stop, Hospital Ahead, Distance boards, Route markers.
- Role in Safety: Helps drivers plan movements calmly. For example, a green board showing “Airport – 5 km” allows safe lane changes.

Traffic Signs

					
Stop	Give way	No parking	No stopping or standing	No entry	One way
					
One way	No way both direction	Right turn prohibited	Left turn prohibited	u turn prohibited	Over taking prohibited
					
Horn prohibited	No entry for cars and motorcycles	Trucks prohibited	Bullock cart prohibited	Pedestrians	Speed limit
					
Y - intersection left	Y - intersection right	Y - intersection	Right hand curve	Left hand curve	Narrow bridge ahead
					
Slippery road	Pedestrian crossing	Falling rocks	School ahead	Cross road	Men at work
					
Public telephone	Petrol pump	Hospital	First aid post	Resting place	Parking both side

Design Specifications of Road Signs in India

According to Indian Road Congress (IRC), road signs must be scientifically designed for visibility, legibility, and uniform interpretation. This is why the specifications of shape, size, color, and lettering are standardized

a) Shape

- Circular → Used for regulatory signs because a circle symbolizes authority and completeness, indicating binding rules.
- Triangular → Used for warnings; its pointed edges create psychological alertness.
- Rectangular → Used for guidance; the stable shape and space efficient, which is ideal for displaying information.
- Octagonal (STOP) → Unique, cannot be confused even if partly covered or damaged and the driver can still recognize it immediately.
- Inverted Triangle (Give Way) → Easily distinguishable, conveys caution and yielding.

b) Color

Color is equally important because the human brain associates colors with emotions and actions:

- Red → Prohibition, danger (Stop, No Entry, Speed Limit).
- White background with red border → Standard for warning signs.
- Black symbols/text → For clarity and contrast.
- Green background with white letters → For route directions and distance boards.
- Blue background with white symbols → For services/facilities (Parking, Hospital).

c) Size

- Size of signs depends on the speed environment, as the drivers must read from a safe distance depending on speed.
- On highways, signs are larger so they can be read 150–200 m away at high speed.
- On urban roads, smaller signs are sufficient since speeds are lower, and less distance is needed for reaction.
- Example: Lettering height is 100 mm for urban roads, 200 mm for highways.

d) Lettering & Symbols

- Must be simple, bold, and legible.
- Decorative fonts are avoided because drivers cannot afford to spend more than a second or two reading.

- Symbols are preferred over text for quick recognition and to assist illiterate/foreign drivers.
- Example: A pedestrian crossing symbol is faster to interpret than the words “Pedestrian Crossing.”

e) Reflectivity & Night Visibility

- IRC mandates use of retro-reflective sheeting so that signs glow when headlights fall on them. As the signs must be visible both day and night
- On urban and expressway routes, LED-lit boards are increasingly used.

Guidelines for Installation of Signs in India

Proper installation ensures signs are noticed and acted upon in time. IRC prescribes strict guidelines:

1. Location

- Warning signs: Before the hazard (≈ 50 m in urban, 150 m in highways).
- Regulatory signs: At or just before the enforcement point.
- Informatory signs: Well in advance to allow safe decisions.

2. Height

- Lower edge of sign should be 2.1 m above ground in cities.
- Higher for rural/highways to avoid obstruction by vehicles or pedestrians.

3. Position

- Always on the left side of the road as per Indian standards (sometimes repeated on right/overhead in multilane roads).

4. Spacing

- Avoid clutter → Too many signs together confuse drivers.
- Logical sequence → Warning first, then Regulation, then Information.

5. Visibility

- Signs must not be hidden by trees, poles, hoardings.
- Must face approaching traffic at the correct angle.

6. Maintenance

- Must be kept clean, upright, and reflective at night.
- Damaged/faded signs must be replaced immediately.

Effect of Improper Placement of Sign

Even the best-designed signs fail if they are not placed correctly. Improper placement can mislead or confuse drivers, creating conditions that directly cause accidents.

- If a sign is placed too close to the hazard, drivers have no time to react, leading to panic braking or swerving. For example, a speed breaker sign put just 5 meters before a hump is practically useless, as the driver cannot slow down in time.
- If placed too far, drivers may forget the warning and fail to act, reducing the effectiveness of the sign.
- When signs are hidden by obstacles such as trees, poles, or advertisements, they become invisible to drivers. Missing a “Curve Ahead” or “Stop” sign can directly result in fatal accidents.
- Similarly, overcrowding of signs in one place leads to information overload. Drivers may get confused, miss the critical instruction, or make hasty, unsafe decisions.

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Beyond their individual categories, signs as a whole serve several safety functions:

- *Enforce Discipline* → Prevent reckless driving through speed limits, parking restrictions, and prohibitions.
- *Provide Early Warning* → Reduce accident risk by alerting drivers to hazards before they encounter them.
- *Guide Drivers* → Reduce stress and confusion, particularly on highways and in unfamiliar areas.
- *Support Law Enforcement* → Regulatory signs give police and courts a basis for penalizing violations.
- *Build Driver Confidence* → Uniform designs across India ensure that drivers can instantly recognize signs, reducing mental strain.

Types Of Road Markings

Road markings are lines, symbols, and patterns painted on the road surface. They complement road signs and signals by giving continuous guidance and information directly on the pavement. The Indian Roads Congress (IRC:35–1997) specifies the standards for road markings in India.

They serve multiple purposes: lane discipline, separation of traffic, pedestrian safety, parking control, and hazard indication.

Classification of Road Markings

1. Longitudinal Markings

These are lines drawn parallel to the direction of traffic flow. They divide lanes, separate opposing streams, and indicate road edges. They are the most common type of road markings and help maintain order in moving traffic.

a) Lane Line Markings

These divide lanes moving in the same direction, guiding vehicles to stay within their designated lanes.

- Examples:
 - Single broken white line → allows lane changing when safe.
 - Solid white line → prohibits lane changing.
- Role in Safety: By clearly separating lanes, they prevent side collisions. For example, solid white lines on urban arterial roads prevent vehicles from weaving dangerously between lanes.

b) Centre Line Markings

Painted along the middle of two-way roads, they separate traffic moving in opposite directions.

- Examples:
 - Broken white line → overtaking allowed when clear.
 - Solid yellow/white line → overtaking prohibited.
 - Double solid yellow lines → absolute no-overtaking rule in both directions.
- Role in Safety: Prevents head-on collisions by reminding drivers to stay on their side. On busy highways, double solid yellow lines are critical against fatal crashes.

c) Edge Line Markings

Continuous lines drawn along the edges of carriageways to mark the usable road boundary.

- Examples: Continuous white edge lines on highways.
- Role in Safety: They act as visual guides during night, rain, or fog, preventing vehicles from drifting off the road. For example, edge lines reduce nighttime run-off accidents on rural roads.

2. Transverse Markings

These are markings placed across the road, mainly used at intersections and pedestrian crossing points. They control stopping positions and right-of-way.

- Examples:
 - STOP line → thick white line before signals or intersections.
 - Give Way line → broken line at junctions with inverted triangle, indicating priority.
 - Zebra crossing → broad white stripes indicating pedestrian right-of-way.
- Role in Safety: They ensure vehicles stop at safe distances, give priority to pedestrians, and reduce crashes at intersections. For example, zebra crossings in front of schools reduce child pedestrian accidents.

3. Hazard Markings

These highlight areas of potential danger and guide vehicles around obstacles or restricted zones.

- Examples:
 - Chevron/divergent markings at road narrowings or medians.
 - Diagonal markings before obstructions or islands.
- Role in Safety: They warn drivers about upcoming hazards and channelize movement, reducing sudden swerves. For example, hazard markings before a median opening prevent abrupt lane changes.

4. Parking Markings

These markings indicate designated parking spaces to organize vehicle standing areas.

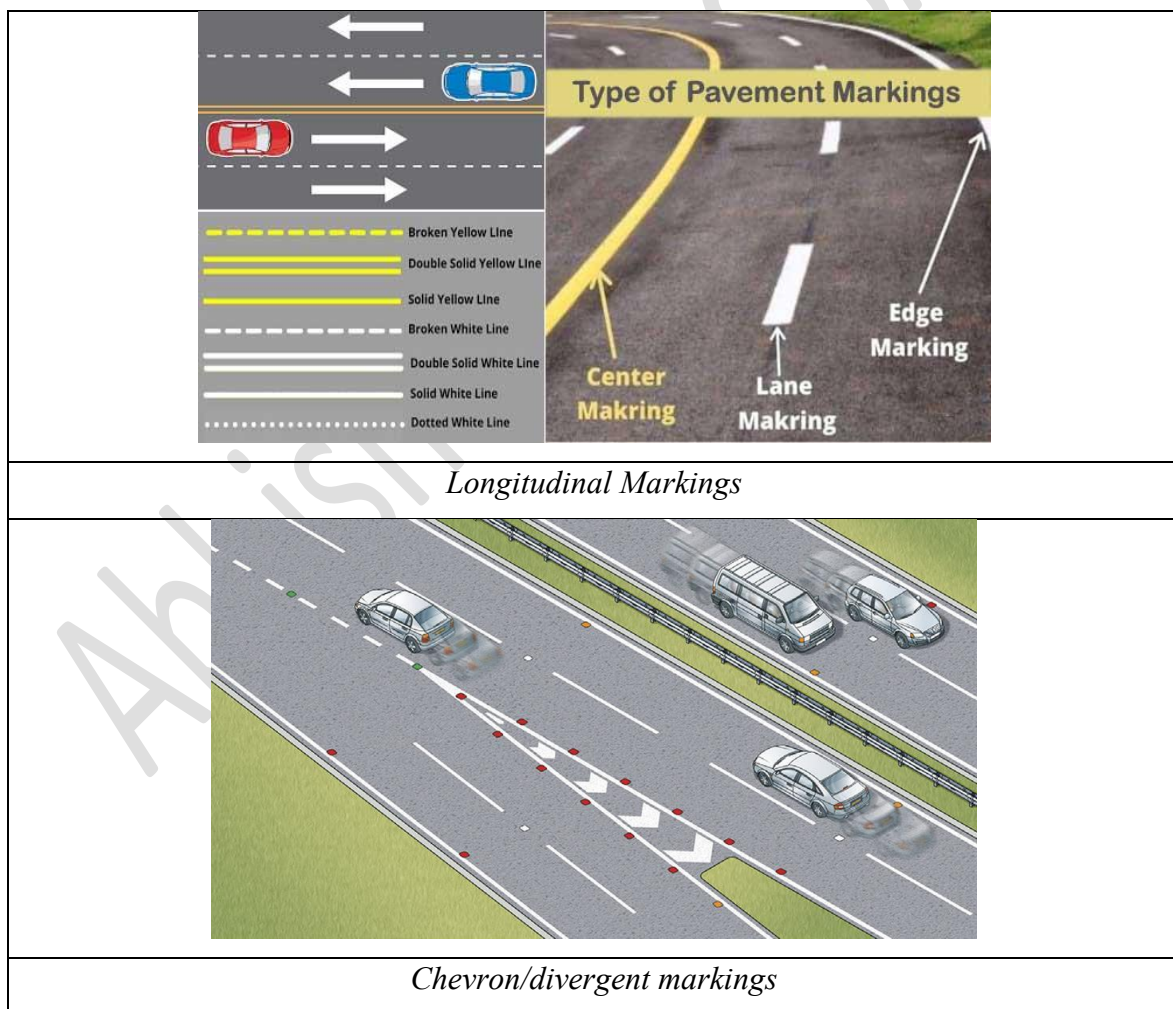
- Example: White or yellow painted rectangles indicating individual parking slots.

- Role in Safety: They reduce random, haphazard parking that blocks visibility and causes congestion. Organized parking ensures smoother traffic and safer pedestrian movements.

7. Special Markings

These are markings meant for specific road users or special situations.

- Examples:
 - Bus lane markings.
 - Cycle lane markings.
 - Speed breaker approaches (triangular or transverse stripes).
- Role in Safety: They protect vulnerable road users like cyclists and pedestrians, and ensure that speed breakers are visible in time to reduce accidents.





Diagonal markings

Design Specifications of Road Markings

The effectiveness of road markings depends on their design, visibility, and durability. Poorly visible or faded markings are almost as bad as no markings at all. Hence, IRC specifies uniform standards.

Specifications

1. Color

- White → Separation of traffic moving in the same direction, lane markings, stop lines, zebra crossings.
- Yellow → Separation of traffic in opposite directions, no-overtaking zones, bus lanes.
- Blue → Parking spaces.
- Red → Special areas (bus bays, cycle crossings).

2. Width & Dimensions

- Centre lines: 100–150 mm wide.
- Stop lines: 300–500 mm wide.
- Pedestrian crossings: Stripes of 500 mm width with 500 mm spacing.
- Edge lines: 150 mm wide.
- These dimensions ensure visibility from a distance.

3. Pattern (Solid vs Broken)

- Broken line → Movement permitted (lane change, overtaking).
- Solid line → Restriction (no crossing).
- Double solid line → Strict prohibition (high-risk zones).

4. *Reflectivity*

- Markings must use thermoplastic paint with reflective glass beads to remain visible at night and in wet conditions.

5. *Placement*

- Markings should be applied uniformly, with proper spacing and orientation.
- At intersections, markings must guide both vehicle movement and pedestrian crossing.

6. *Durability*

- Materials like thermoplastic paint last longer than ordinary paint.
- Maintenance is essential to prevent fading.

Role of Road Markings in Safety

Road markings act as the continuous guidance system for road users. Unlike signs (which are seen only at intervals), markings provide real-time, on-road instructions. Their role in safety is immense, particularly in ensuring lane discipline, pedestrian protection, and nighttime driving guidance.

Contributions to Safety

1. *Lane Discipline and Separation of Traffic*

- Prevents head-on and side collisions by guiding vehicles within lanes.

2. *Pedestrian Protection*

- Zebra crossings and pedestrian refuge areas safeguard pedestrians, especially in cities and school zones.

3. *Intersection Safety*

- STOP lines, Give Way lines, and directional arrows ensure safe and predictable movement at junctions.

4. *Hazard Indication*

- Chevron and diagonal markings warn of obstructions, bottlenecks, and merging points.

5. *Night-time and Adverse Weather Safety*

- Reflective edge lines and thermoplastic paint markings improve visibility during fog, rain, or darkness.

6. *Parking Management*

- Organized parking bays prevent random vehicle stopping and improve pedestrian sightlines.

7. *Guidance for Turning Movements*

- Arrows painted on lanes direct vehicles for left turn, right turn, or straight movement, reducing confusion and side crashes at intersections.

8. *Speed Management*

- Markings such as rumble strips, transverse bars, or triangular approach markings near speed breakers alert drivers to slow down.

9. *Support for Enforcement*

- Clearly marked STOP lines or bus lane markings allow traffic police to monitor and fine violators, indirectly improving discipline and safety.

10. *Psychological Comfort to Drivers*

- Continuous markings act as visual guides, particularly for long-distance drivers on highways, reducing fatigue and stress.